CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Banking (Recurring Deposit Accounts)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Interest and maturity value

1. A man deposits ₹500 every month for 18 months in a recurring deposit account and earns ₹570 as interest. Which expression gives the maturity amount? (a) ₹$(500 \times 18 + 570)$ (b) ₹$(500 \times 19 + 570)$ (c) ₹$(500 \times 18 \times 19 + 570)$ (d) ₹$(500 \times 9 \times 19 + 570)$ ICSE 2025, 1 mark

SOLUTION

Maturity amount = total money deposited + interest.

Total deposited $= 500 \times 18$.

Answer: (a)

2. A boy deposits ₹800 every month for 6 months in a recurring deposit account and receives ₹4884 on maturity. How much interest does he earn? (a) ₹84 (b) ₹42 (c) ₹24 (d) ₹284 ICSE 2023, 1 mark

SOLUTION

Total deposited $= 800 \times 6 = 4800$

Interest $= 4884 - 4800 = 84$

Answer: (a)

3. A girl deposits ₹600 a month in a recurring deposit account for $2\frac{1}{2}$ years at 10% per annum. Find the maturity value. ICSE 2018, 3 marks

SOLUTION

$n = 30$ months. Interest $= 600 \times \frac{30 \times 31}{2 \times 12} \times \frac{10}{100} = 600 \times 38.75 \times 0.1 = 2325$

Total deposited $= 600 \times 30 = 18000$

Answer: maturity value = ₹20,325

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2 Finding the rate of interest

4. A woman deposits ₹250 a month in a recurring deposit account for 3 years and receives ₹10,110 on maturity. Find (a) the rate of interest (b) how much more interest she would earn by depositing ₹50 more each month, at the same rate and for the same time. ICSE 2025, 4 marks

SOLUTION

(a) $P = 250$, $n = 36$. Total deposited $= 250 \times 36 = 9000$, so the interest is $10110 - 9000 = 1110$.

$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$: $1110 = 250 \times \frac{36 \times 37}{24} \times \frac{r}{100} = 138.75\,r$

$r = \frac{1110}{138.75} = 8$. **The rate is 8% per annum.**

(b) The extra ₹50 a month earns $50 \times \frac{36 \times 37}{24} \times \frac{8}{100} = 50 \times 55.5 \times 0.08 = 222$.

Answer: (a) 8% per annum (b) ₹222 more

5. A man deposits ₹1000 every month in a recurring deposit account for 2 years and receives ₹26,000 on maturity. Find (a) the total interest he earns (b) the rate of interest. ICSE 2023, 4 marks

SOLUTION

(a) Total deposited $= 1000 \times 24 = 24000$, so the interest is $26000 - 24000 = 2000$.

(b) $I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$: $2000 = 1000 \times \frac{24 \times 25}{24} \times \frac{r}{100} = 250r$

$r = 8$

Answer: (a) ₹2000 (b) 8% per annum

6. A man deposits ₹750 a month in a recurring deposit account for 2 years and receives ₹19,125 on maturity. Find the rate of interest. ICSE 2020, 3 marks

SOLUTION

Total deposited $= 750 \times 24 = 18000$, so the interest is $19125 - 18000 = 1125$.

$1125 = 750 \times \frac{24 \times 25}{2 \times 12} \times \frac{r}{100} = 187.5\,r$

$r = 6$

Answer: 6% per annum

3 Finding the time

7. A man deposits ₹2000 a month in a recurring deposit account that pays 8% per annum. He receives ₹1040 as interest on maturity. For how many years was the account held? ICSE 2024, 3 marks

SOLUTION

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} : 1040 = 2000 \times \frac{n(n+1)}{24} \times \frac{8}{100}$$

$1040 = \frac{20}{3} n(n+1)$, so $n(n+1) = 156 = 12 \times 13$

$n = 12$ months

Answer: 1 year

8. A recurring deposit account of ₹1000 a month at 10% per annum earns ₹5550 as interest on maturity. For how long was the account held? ICSE 2018, 3 marks

SOLUTION

$5550 = 1000 \times \frac{n(n+1)}{24} \times \frac{10}{100}$, so $n(n+1) = 1332 = 36 \times 37$

$n = 36$ months

Answer: 3 years

4 Finding the monthly deposit

9. A recurring deposit account is held for 20 months at 9% per annum, and the interest received on maturity is ₹441. Find the amount deposited each month. ICSE 2019, 3 marks

SOLUTION

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} : 441 = P \times \frac{20 \times 21}{24} \times \frac{9}{100} = 1.575P$$

$$P = \frac{441}{1.575} = 280$$

Answer: ₹280 per month

10. A recurring deposit account is held for 3 years at 7.5% per annum, and the interest on maturity is ₹8325. Find (a) the monthly deposit (b) the maturity value. ICSE 2017, 3 marks

SOLUTION

(a) $8325 = P \times \frac{36 \times 37}{2 \times 12} \times \frac{7.5}{100} = P \times 55.5 \times 0.075 = 4.1625P$, so $P = 2000$.

(b) Total deposited $= 2000 \times 36 = 72000$. Maturity value $= 72000 + 8325$.

Answer: (a) ₹2000 (b) ₹80,325

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